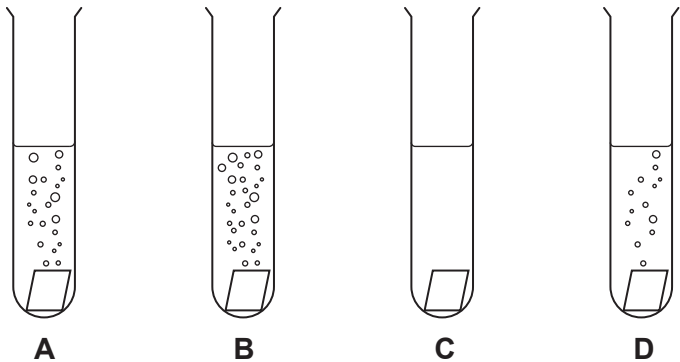


3. Nathan and Simon were investigating the reactivity of metals.

(a) In one experiment, they investigated the effect of adding dilute hydrochloric acid to four different metals, **A**, **B**, **C** and **D**.

The diagram shows the observations made.



(i) They used the same **volume** of acid in each tube. Give **two other** ways they controlled the dilute hydrochloric acid to ensure a fair test. [2]

- 1.
- 2.

(ii) Use the observations to list the metals in order of their reactivity. Give the reason for your choice. [2]

Most reactive

.....

.....

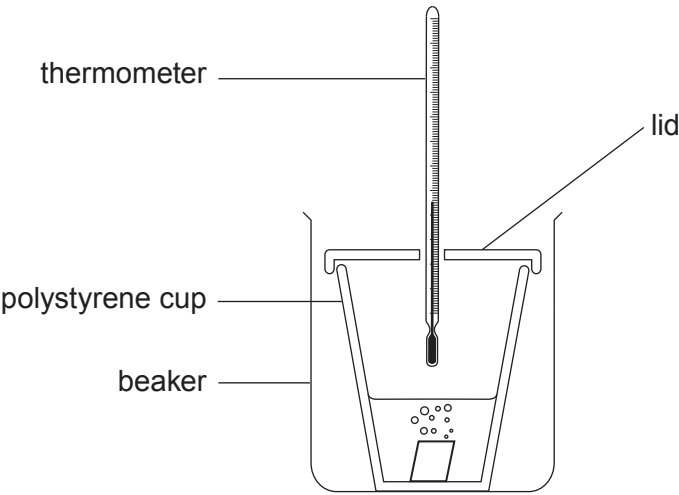
Least reactive

Reason

.....



- (b) In a second experiment, Nathan and Simon used the following apparatus to investigate the temperature rise when metals **A**, **B**, **C** and **D** were added to hydrochloric acid.



- (i) Identify the error in the way the apparatus is set up and state how this would affect the results collected. [2]

Error

Effect

.....

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(ii) Once they had set up their apparatus correctly, they obtained the following results.

Metal	Temperature rise (°C)			
	Test 1	Test 2	Test 3	Mean
A	22	21		22
B	40	38	36	38
C	0	0	0	0
D	15	9	13	14

I. Identify the anomalous result from metals **B**, **C** and **D**. [1]

Metal Test

II. Calculate the missing result for metal **A**. [1]

Temperature rise for test 3 = °C

III. Zinc is more reactive than metals **C** and **D** but less reactive than metals **A** and **B**. Use this information to predict the temperature rise that would be expected for zinc. [1]

Temperature rise for zinc °C



- (c) The reaction of a metal with an acid is summarised by the following equation.



- (i) Complete the equation for the reaction between magnesium and hydrochloric acid by
- giving the formula of the salt magnesium chloride,
 - balancing the equation.

[2]



- (ii) Give the name of the salt formed when magnesium reacts with nitric acid.

[1]

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